

Addressing 2028 Capex and AI Platform Value Shift

Jeff Pu, CFA

SFC CE No. BNO719

jeffpu@gfgroup.com.hk

Henry Huang

SFC CE No. BYA749

henryhuang@gfgroup.com.hk

What's New?

We provided our view on U.S. CSP capex in 2027/2028, the key debate in the market, and preferred sectors. In sum, we forecast U.S., CSP capex to be \$1.2tn/\$1.4tn, or 45% YoY/20% YoY, supported by healthy balance sheet, yet it suggests a deceleration in 2028. We remain largely positive on AI trends and prefer GPU/CPU/Optics. On the other hand, we are conservative on commodities (i.e., memories).

Comments

U.S. CSP capex to be 85%/45% YoY in 2026/2027: Most CSPs raised 2026 capex guidance in recent earnings except Microsoft - Amazon to \$220bn from \$200bn; Google to \$195–205bn from \$180–190bn; Meta from \$125–145bn to \$130–145bn; Oracle guided FY27 capex to \$90–95bn, up from \$55.7bn in FY26 (+62% YoY). CoWoS capacity also reflects the strong demand, where we revised up TSMC and OSAT capacity in our earlier reports. As such, we now forecast Top 5 CSPs capex of \$823bn/1190bn in 2026E/2027E, or +85%/45% YoY.

U.S. CSP debt issuance up to \$421bn by 2028, but manageable: Given capex expands further by +85%/+45% YoY in 2026/2027, we forecast Top 5 CSPs' debt issuance to reach \$257bn/\$419bn in 2026/2027, or 31%/35% of total capex. Assuming debt issuance to grow in 2028/2029 at a mild pace, we believe U.S. CSPs' balance sheet (except Oracle) to remain very healthy, due to net leverage (EBITDA basis) of only 0.5 and the net debt-to-equity ratio of only 18% by 2029.

U.S. CSP capex moderates to +20% YoY in 2028: Despite healthy balance sheet for most CSPs, we expect most to safeguard balance sheets beyond 2027 amid widening spreads and FCF profiles, with the focus shifting to returns. Together with a swing factor of competition from Chinese LLMs, we expect U.S. CSPs to decelerate capex expansion. We now forecast Top 5 CSPs capex of \$1,428bn/1,642bn in 2028/2029, or +20%/15% YoY, down from +45% in 2027.

Nvidia platform memory cuts: Per our *July 2's report of Managing Memory Cost*, we view LPDDR5x reductions as essential for lowering cost, driving Vera CPU SOCAMM down from the very original 192GB to 96GB in 2H26, with selective SKUs moving to 64GB by 1Q27. For Rubin Ultra, we believe that the major Korean HBM supplier has finalized the 8-Hi design. As a result, Rubin Ultra steps down from 384GB (HBM4E-12Hi) to 192GB (HBM4) or 256 GB (HBM4E) on 8-Hi, while 12-Hi options of 288GB (HBM4) and 384GB (HBM4E) stay in the lineup. We see CSPs—not GPU vendors or supply constraints—as the primary driver of this de-spec. Competitors including AMD's Meta-custom MI450 and Google's TPU (Humufish, Triggerfish) have already shifted to 8-Hi.

Scale-up domain to increase by NVL576 to compensate lower memory: With lower memory spec, we expect Nvidia to increase the scale-up domain by optics to enhance performance. According to Nvidia's GTC 2026, the optical scale-up fabric allows Nvidia to expand Rubin Ultra to an NVL576 scale-up. Regarding CPO and NPO which are being parallelly developed, we expect NPO to be a high reliability scaling path for NPO. We now expect its scale-up to adopt NPO in Rubin Ultra and Feynman platform and estimate OE shipments of Nvidia's platform to be 5.8m/8.1m in 2027/2028.

Recommendations:

- 1) **Nvidia GPU:** NVDA (Buy), Hon Hai (2317 TT), SMCi (Buy)
- 2) **CPU:** INTC (Buy), AMD (Buy), ASE (3711 TT, ASX, Buy)
- 3) **Optics NPO:** LITE (Buy), MRVL (Buy), SMTC (Buy), TSEM (Buy), Bizlink (3665 TT Buy), Browave (3163 TT)

Risks: 1) AI demand slowdown, 2) infrastructure deployment, 3) supply constraints, 4) competition.

U.S. CSP Capex to be \$1.2tn/1.4tn in 2027E/2028E

CSPs raised capex guidance in recent earnings calls

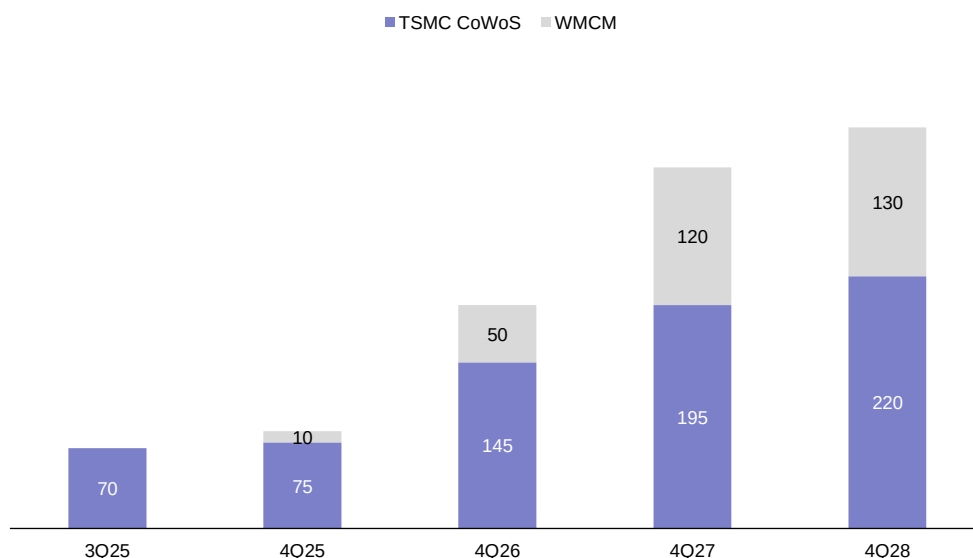
During the recent earnings season, most CSPs raised capex guidance, with Microsoft as the exception. Amazon raised 2026 capex guidance from \$200bn to \$220bn, as mgmt indicated that most 2027 AWS compute capacity has already been committed, with substantial 2028 capacity also locked in, underscoring supply constraints; Google lifted 2026 capex guidance from \$180–190bn to \$195–205bn, with quarterly free cash flow turning negative, reflecting robust AI infrastructure investment driven by strong cloud demand, where the backlog expanded to \$514bn from \$462bn in 1Q26; Meta raised the lower end of 2026 capex guidance by \$5bn, from \$125–145bn to \$130–145bn, where mgmt emphasized that the priority is to deploy AI compute capacity faster throughout 2026–2027; Oracle also guided FY27 capex to \$90–95bn, up from \$55.7bn in FY26 (+62% YoY), alongside an additional \$40bn in debt and equity financing to support the expansion, reflecting its \$638bn RPO.

CoWoS capacity to reflect hyperscalers' demand

Following our earlier raise in May Edition in which we revised up TSMC's CoWoS capacity from 175KPM to 195KPM by end-2027, the Street has moved up to our numbers.

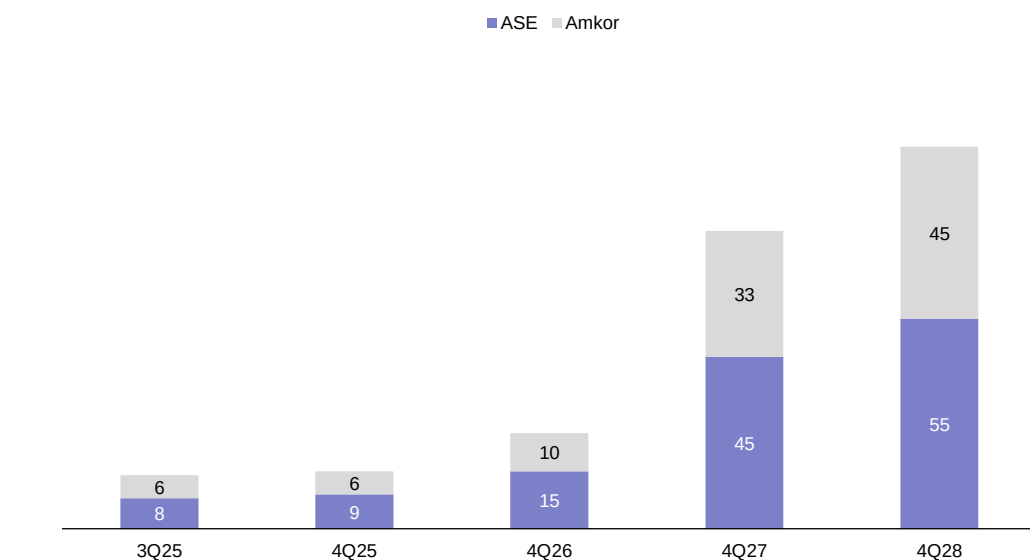
For OSAT, following our earlier upward revision for Amkor in June Edition in which we revise up Amkor's capacity to 15KPM/25KPM/40KPM by end of 2026/2027/2028, we recently revised up ASE's CoWoS capacity 15KPM/45KPM/55KPM by end 2026/2027/2028 with our July 3's report.

Figure 1: TSMC's quarterly CoWoS + WMCM capacity (kwpm) trend



Sources: Company data, IDC, GF Securities (Hong Kong) Brokerage

Figure 2: Amkor and ASE's quarterly CoWoS (kwpm) forecast



Sources: Company data, IDC, GF Securities (Hong Kong) Brokerage

Maintain CSPs capex of +85%/45% YoY in 2026/2027

As said, CSPs are investing aggressively to meet AI infrastructure needs, while record revenue and backlog growth of cloud business reinforce the conviction to revise up capex. Based on the guidance in recent CSPs earnings call, we now forecast Top 5 CSPs capex of \$823bn/1190bn in 2026E/2027E, representing +85%/45% YoY.

Figure 3: US Top-5 CSPs capex (USD bn)

	2024	2025	2026E	2027E	2028E	2029E
MSFT	76	118	195	253	304	349
AMZN	83	132	220	308	370	425
Google	53	91	205	322	386	444
Meta	39	69	141	206	247	284
Oracle	11	35	67	101	121	139
Top 5	261	446	823	1,190	1,428	1,642
YoY	62%	71%	85%	45%	20%	15%

Sources: Company data, GF Securities (Hong Kong) Brokerage

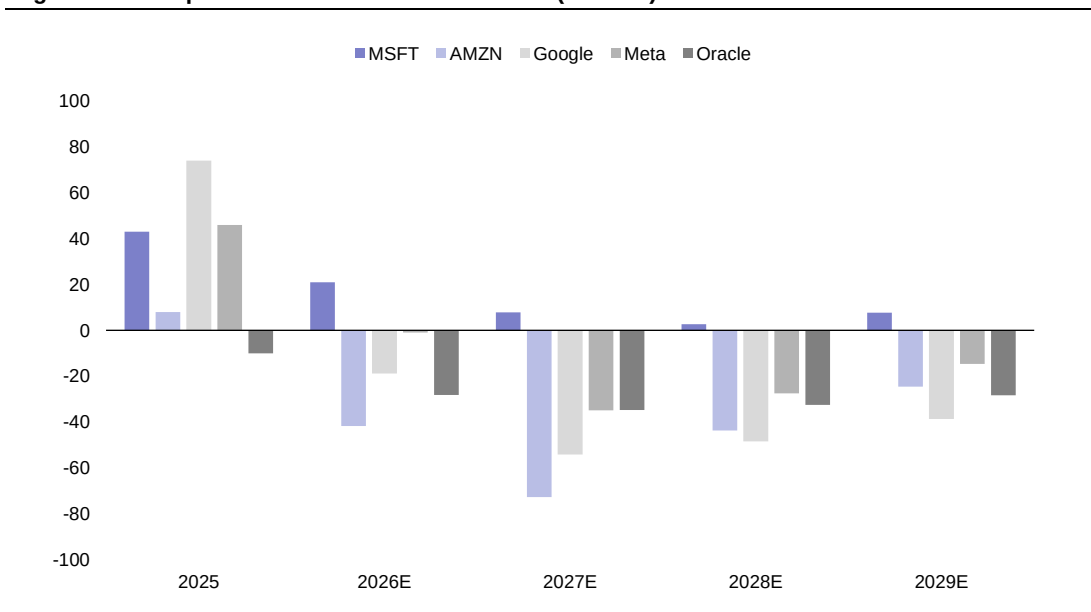
U.S. CSP Debt Issuance Up to \$421bn by 2028, But Not a Problem

Hyperscalers shift to debt-financed capex buildout

CSPs used to rely on CFO to cover capex, while free cash flow profiles are undergoing significant pressure as AI infrastructure investments surge. Google's 2Q26 FCF turned negative, reaching -\$5.9bn; Amazon's TTM FCF for 2Q26 dropped to -\$7.6bn, vs. \$18.2bn in 2Q25 (-142% YoY); Meta's 2Q26 FCF plummeted 91% YoY to \$784mn, likely turning negative in 2H26; Oracle's FCF fell to -\$23bn in FY26.

Against this backdrop, Hyperscalers turn to debt financing for their massive capex. In 2025, Top 5 CSPs' total debt issuance reached \$108bn, representing 24% of total capex, while their FCF remained positive excluding Oracle, suggesting that pre-funding for capital outlay had already begun. Given capex expands further by +85%/+45% YoY over 2026-2027 and FCF turns negative for CSPs except Microsoft, we now forecast Top 5 CSPs' debt issuance to reach \$257bn/\$419bn in 2026/2027, or 31%/35% of total capex.

Figure 4: US Top-5 CSPs free cash flow forecast (USD bn)



Sources: Company data, GF Securities (Hong Kong) Brokerage

Hyperscalers balance sheet to remain healthy into 2029

Based on our forecast into 2029, excluding off-balance sheet obligations, Top 5 CSPs' gross leverage (EBITDA basis) is expected to be 1.1, net leverage (EBITDA basis) slightly up to 0.5, and the net debt-to-equity ratio of only 18%, while CFO-to-debt decreases to 1.0 and interest expense coverage still up to 14. In theory, hyperscalers' intact balance sheet metrics and robust 14x interest coverage should afford them ample runway to issue additional debt for financing capex.

Figure 5: U.S. Top-5 CSP leverage and solvency metrics

	2025	2026E	2027E	2028E	2029E
Key assumptions					
Revenue YoY	15%	18%	17%	16%	14%
EBITDA YoY	22%	21%	24%	24%	18%
Gross Debt (bn)	297	537	889	1,236	1,574
Net Debt (bn)	100	220	442	631	766
Key ratios					
Gross leverage (EBIT basis)	0.6	1.0	1.3	1.5	1.5
Gross leverage (EBITDA basis)	0.5	0.7	0.9	1.0	1.1
Net leverage (EBIT basis)	0.2	0.4	0.6	0.7	0.8
Net leverage (EBITDA basis)	0.2	0.3	0.4	0.5	0.5
Net debt / equity	7%	11%	17%	19%	18%
CFO / debt	2.0	1.4	1.1	1.0	1.0
Interest expense coverage	43	26	19	17	16

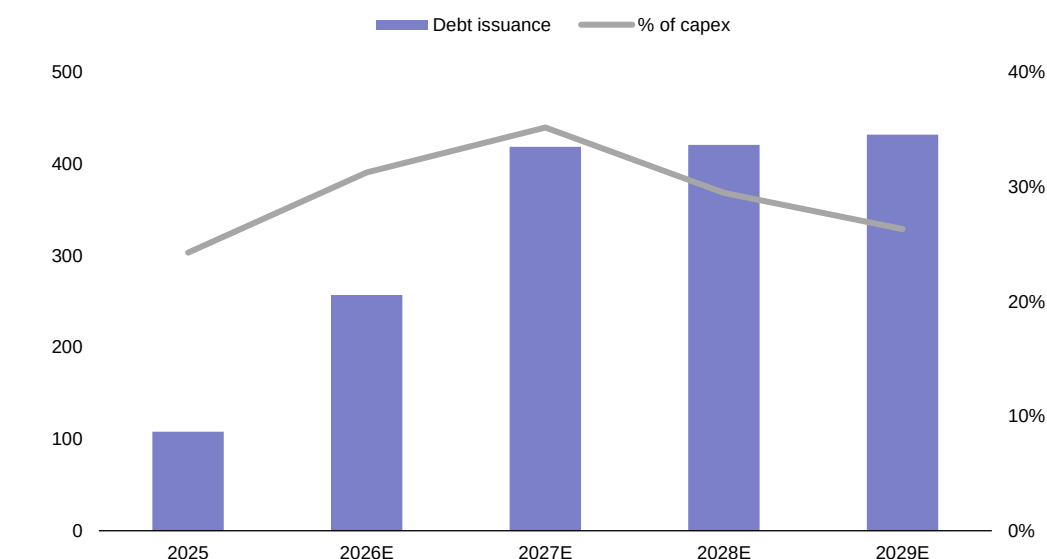
Sources: Company data, GF Securities (Hong Kong) Brokerage

Debt market capacity, widening spreads are concerns but manageable

For further debt issuance, the market concerns about: **1) debt market capacity constraints**, issuer concentration and market saturation; **2) widening spreads**, surging bond supply and lengthening maturities are driving up borrowing spreads over risk-free rates, with median spreads on 2-4 year bonds have widened to 40 bps from 30 bps in 2025, 5-7 year to 60 bps from 50 bps, and long 20+ year to 118 bps from 108 bps. As such, we now expect Top 5 CSPs' debt issuance to be \$421bn/\$432bn in 2028–2029, slightly up from \$419bn in 2027, assuming the growing new profit on cloud revenue.

However, based on our balance sheet metrics analysis which considers growing debt issuance and interest spreads, U.S. CSPs except Oracle could remain intact and manageable.

Figure 6: Hyperscalers debt issuance (USD bn)



Sources: Company data, GF Securities (Hong Kong) Brokerage

We forecast U.S. CSP capex to +20% YoY in 2028, considering balance sheet and AI model competition

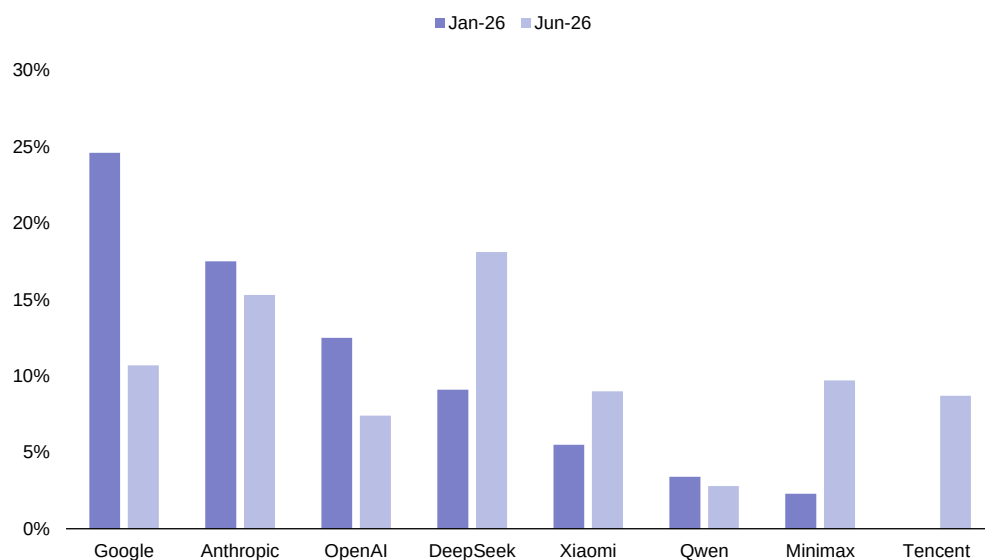
We expect most CSPs to safeguard balance sheets beyond 2027 amid rising interest rates and FCF profiles, especially when CSPs shifts focus to profitability and returns.

Moreover, uncertainty of competition from Chinese LLMs is also a watch. According to Openrouter, Chinese LLMs are gaining market share from US counterparts. Over January to June in 2026, DeepSeek nearly doubled its monthly token volume share from 9.1% to 18.1%, Xiaomi rose to 9.0% from 5.5%, MiniMax surged to 9.7% from 2.3%, and Tencent also captured a sizeable ~8.0% share, while Anthropic's contracted from 17.5% to 15.3%, OpenAI slipped from 12.5% to 7.4%, and Google dropped by more than half to 10.7% from 24.6%, which is mostly driven by higher performance/cost of Chinese LLMs.

Among frontier LLMs, the input/output token price per million tokens of DeepSeek V4 to be \$0.084/\$0.168, Kimi K3 to be \$2.9/\$14, MiniMax M3 to be \$0.24/0.96 (vs. Claude Opus 5 of \$5/\$25, GPT 5.6 Sol of \$5/\$30, and Gemini 3.6 of \$1.5/\$7.5). With ~50% of Top 5 US CSPs' RPOs concentrated in OpenAI and Anthropic, the potential token price competition driven by Chinese models could lead to pressures on US LLM margins. This dynamic casts doubt on both the standalone monetization of LLMs and the ability of CSPs to convert their massive capex into revenue.

Net, we expect balance sheet discipline and potential competition from Chinese LLMs to lead hyperscalers to decelerate capex expansion despite intact balance sheet metrics. We now forecast Top 5 CSPs capex of \$1428bn/1642bn in 2028/2029, or +20%/15% YoY, down from +45% in 2027.

Figure 7: Share of monthly token volume, Jan-26 vs. June-26



Sources: Openrouter, GF Securities (Hong Kong) Brokerage

Figure 8: Frontier LLMs token price (USD\$ per million tokens)

Region	Model	Input price	Output price
China	DeepSeek V4	0.084	0.168
	Kimi K3	2.9	14
	MiniMax M3	0.24	0.96
US	Claude Opus 5	5	25
	GPT 5.6 Sol	5	30
	Gemini 3.6	1.5	7.5

Sources: Openrouter, GF Securities (Hong Kong) Brokerage

Nvidia Platform's Shift in Content

Nvidia's rollout of Vera Rubin intact

Nvidia CEO Jensen Huang shared initial public updates on Vera Rubin progress during his GTC Taipei keynote on June 1, 2026. He confirmed Vera Rubin silicon had completed TSMC fabrication runs and returned to Nvidia for validation, on track for mass production and volume shipments in 2H26. The company suggests Vera Rubin is an extreme co-designed platform built around seven new chips. Distinguishing the architecture from Blackwell, the CEO stated Vera Rubin is optimized for agentic AI workloads. Spec-wise, we expect the **Rubin to be only 2300W on mass volume version, vs. sampling version of 1800W.**

According to Nvidia's NDR on July 10, the CEO addressed emerging market concerns regarding potential technical risks and timeline slippage for Vera Rubin. He specifically rebutted rumours that Rubin Ultra's launch could be pushed into 2028, reiterating adherence to the existing internal schedule.

Figure 9: Nvidia's GPU roadmap till Rubin

	B200	B300	Rubin
Launch/MP	4Q24	3Q25	Sept-26
Architecture	2-die	2-die	2-die + 2xI/O
Front-end node	N4P	N4P	N3P
Back-end process	CoWoS-L	CoWoS-L	CoWoS-L
Transistor (bn)	208	208	
Interconnect	NVLink 1,800 GB/s	NVLink 1,800 GB/s	NVLink 3,600 GB/s
Memory (GB)	192GB HBM3E	288GB HBM3E 12hi x 8	288GB HBM4 12hi x 8
Memory Bandwidth (TB/s)	8	8	20.5
TDP	1000W	1000W	2,300W
Networking	400Gb CX7	800Gb CX8	Dual-800Gb CX9

Sources: Company GTC 2026, GF Securities (Hong Kong) Brokerage.

Reiterate our analysis on memory cost to VR200 NVL72 on de-spec

Per our July 2's report on *Supply Chain Taking Actions to Manage Memory Cost*, our analysis indicates that reducing LPDDR5x capacity is a necessary shift to manage costs, since this component initially represents approximately 61% of the rack's total memory and storage budget. Thus, we expect an aggressive capacity reduction for LPDDR5x by half, with a potential further cut to one-fourth of the original spec under an extreme scenario. This adjustment will effectively lower LPDDR5x rack costs from \$774k to \$293k, instead of escalating to \$1.2m.

Without these capacity adjustments for system BOM optimization, memory and storage costs would escalate to \$2.1mn, or 29% of the total BOM (up from 19%). That said, by implementing the cut to 768GB or even 384GB per CPU, costs are successfully managed back to the ~20% level.

Figure 10: VR200 NVL72 Cost Analysis by 4Q26

Cost Analysis	Original Spec	De-spec/main Ver.	Extreme Scenario	w/o Spec Change
Price per unit				
HBM4 (\$/Gb)	2.0	4.0	4.0	4.0
LPDDR5x (\$/GB)	14.0	21.2	21.2	21.2
SSD (\$/GB)	0.27	0.45	0.45	0.45
Content per rack (TB)				
HBM4 (288GB/GPU)	20.7	20.7	20.7	20.7
LPDDR5x	55.3	27.6	13.8	55.3
Capacity per CPU	1536 GB	768 GB	384 GB	1536 GB
SSD (32.7TB/Compute tray)	589	589	589	589
Cost per rack (\$k)				
HBM4	332	664	664	664
LPDDR5x	774	586	293	1172
SSD	159	265	265	265
Total (\$k)	1,265	1,515	1,221	2,101
Cost increase per rack (\$k)		250	-43	836
BOM Analysis	Original Spec	4Q26E Ver.	Extreme Scenario	w/o Spec Change
Memory & Storage Cost	1,265	1,515	1,221	2,101
Rack BOM (USD k)	6,500	6,750	6,457	7,336
Memory % of BOM	19%	22%	19%	29%

Sources: Company Data, GF Securities (Hong Kong) Brokerage.

Rubin Ultra moves to HBM4E 8-Hi

According to news report on Aug 3 about SemiAnalysis notes that Nvidia's Rubin Ultra has been lowered, with HBM reduced from 12-Hi/384GB to 8-Hi/192GB and power from ~2300W to ~1800W.

For us, we believe that Rubin Ultra is evaluating an HBM4/HBM4E 8-Hi, depending on HBM4's speed improvement on per pin speed. We believe that the major Korean HBM supplier has finalized the 8-Hi design. For Rubin Ultra's design trend, under the Rubin Ultra two-die, eight-stack configuration, total HBM capacity steps down from 512 GB using HBM4E-16Hi to 384 GB on HBM4E-12Hi, and then to 192 GB (HBM4) or 256 GB (HBM4E) for the 8-Hi generation, with corresponding 12-Hi capacities of 288 GB (HBM4) and 384 GB (HBM4E).

We believe the de-spec is driven by CSP customers, rather than GPU makers, and believe this will push server vendors to recalibrate system BOMs, which addressed in later session. We believe Nvidia's competitors such as AMD (Meta's custom MI450) and Google TPU (Humufish, Triggerfish) have moved to 8-Hi.

Figure 11: Potential Rubin Ultra HBM spec map

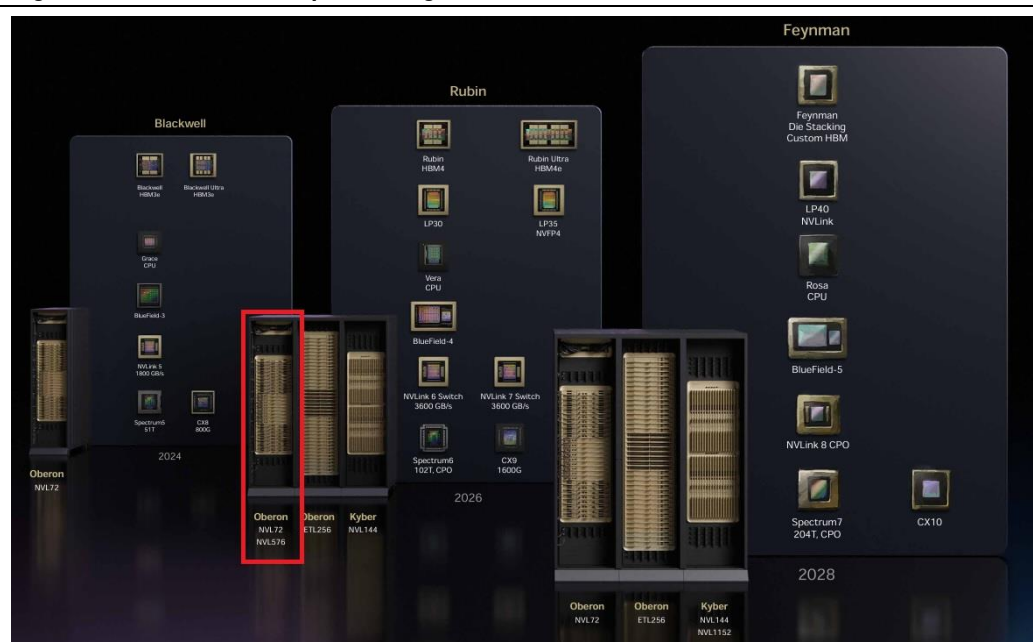
	HBM4	HBM4E
8 Hi	192GB	256GB
12 Hi	288GB	384GB

Sources: Company data, GF Securities (Hong Kong) Brokerage

Scale-up domain to increase by NVL576 to compensate lower memory spec

With lower memory spec, we expect Nvidia to increase the scale-up domain by optics to enhance performance. According to Nvidia's roadmap shown on GTC 2026, the optical scale-up fabric allows Nvidia to expand Rubin Ultra to an NVL576 scale-up, enabling eight interconnected 72-GPU racks and improving efficiency for mixture-of-experts architectures.

Figure 12: Nvidia's roadmap including NVL576



Sources: Computex 2026, GF Securities (Hong Kong) Brokerage.

Within Nvidia's NVL576 and broader CSPs' AI infrastructure plans, CPO and NPO are being parallelly developed to expand GPU/ASIC interconnect bandwidth: CPO is the ideal long-term design for lower power and latency yet faces headwinds from complicated advanced packaging, and yield risks, while NPO could deliver most short-reach optical performance gains with far better manufacturability and deployment flexibility, making NPO a reliable near-term scaling path and CPO the ultimate end-state architecture.

Nvidia's NPO OE to be 5.8m/8.1m in 2027/2028

For Nvidia, we now expect its scale up to adopt NPO in Rubin Ultra and Feynman platform and estimate OE shipments of Nvidia's platform to be 5.8m/8.1m in 2027/2028, based on 20% NPO adoption in 2027. Coupled with other CSP's platform, we estimate total OE market to be 11m/40m in 2027/2028.

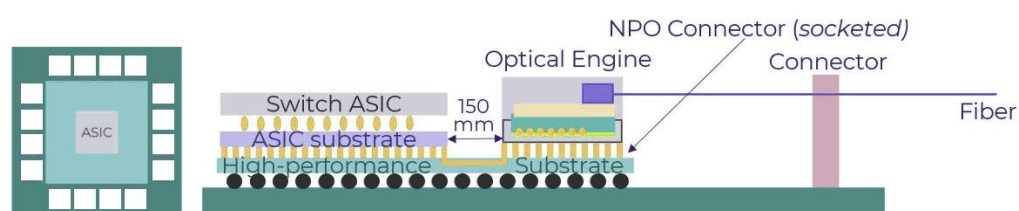
TIA/Driver to benefit from NPO

NPO partially removes module-side DSP and shifts signal conditioning and equalization to system designs, greatly lifting the value of TIAs and laser drivers that take over part of traditional DSP functions; we estimate each 3.2T optical engine contains tens of dollars worth of combined TIA and driver chips, a clear improvement over standard pluggable optics.

A key architectural shift under NPO is the partial removal of DSP functionality at the module level, with signal conditioning and equalization increasingly redistributed into system-level design. This significantly elevates the importance of TIA (transimpedance amplifier) and laser driver components, which effectively replace a portion of traditional DSP value. We estimate that each 3.2T optical engine will embed high tens of dollars of combined TIA and driver content, representing a meaningful step-up versus traditional pluggable optics architectures.

From a competitive standpoint, we expect Marvell to remain the primary beneficiary of both NPO OE given its stronger positioning high speed TIA/Driver. Semtech is well positioned as a secondary beneficiary through its exposure to high-speed analog signal chain components, particularly TIAs and laser drivers, which scale directly with optical engine proliferation under NPO. We forecast 11.3mn/40mn NPO OE shipments in CY27/28. Based on our estimated front-end content per OE and a 30% market share assumption for Semtech, we estimate its NPO revenue of \$205/926mn.

Figure 13: Semtech's Serdes IP product portfolio



Sources: Company data, GF Securities (Hong Kong) Brokerage

Figure 14: Nvidia's NPO unit forecast

	2026	2027	2028
Market for Nvidia's scale up NPO(k units)			
NVL72	75	90	60
NVL144 (6 NVSwitch blades)	0	0	35
Number NVSwitch trays/blades			
NVL72	9	9	9
NVL144 (6 NVSwitch blades)			12
Adoption of NPO			
NVL72	0%	20%	30%
NVL144 (6 NVSwitch blades)	0%	0%	15%
NPO NVSwitch trays/blades (k)			
NVL72	0	162	162
NVL144 (6 NVSwitch blades)	0	0	63
Nvidia NPO OE(k)			
Shipments (m units)		5.8	8.1
YoY			39%
Nvidia's TIA/Driver (k,USD)			
TIA & Driver ASP per OE (USD)		65	65
TAM for TIA/Driver (m USD)		379	527
TAM yoy			39%
Market for CSP's scale up NPO(m units)			
OE shipment (m units)		5.5	32
TIA & Driver ASP per OE (USD)		80	80
TAM for TIA/Driver (m USD)		440	2,560
TAM yoy			482%
Total			
Shipments (m units)		11.3	40
TAM for TIA/Driver (m USD)		819	3,087
TAM yoy			277%

Sources: Company data, GF Securities (Hong Kong) Brokerage

Risks

1) AI demand slowdown, 2) infrastructure deployment, 3) supply constraints, 4) competition.

Rating definitions Benchmark: Hang Seng Index (Hong Kong)

Company ratings	Buy	Stock expected to outperform benchmark by more than 10%
	Hold	Expected stock relative performance ranges between -10% and 10%
	Underperform	Stock expected to underperform benchmark by more than 10%

Sector ratings	Positive	Sector expected to outperform benchmark by more than 10%
	Neutral	Expected sector relative performance ranges between -10% and 10%
	Cautious	Sector expected to underperform benchmark by more than 10%

Hong Kong Company	GF Securities (Hong Kong) Brokerage Limited
Address	27/F, GF Tower, 81 Lockhart Road, Wan Chai, Hong Kong
Telephone	(852) 37191111
Email	evanlee@gfgroup.com.hk

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